

Python Tsunami

An Introduction to Python

Who are we?

Center for Health Data Science



Center for Health Data Science (HeaDS)

Section for Health Data Science and Artificial Intelligence (HDSAI)

The mission of HeaDS is to strengthen health data science research at SUND:

- Visible hub for Health Data Science
- Research within the field of HDS
- SUND Data Lab



<https://heads.ku.dk>



COURSES & WORKSHOPS

Data Science skills, Tools
and HDS Topics

COMMISSIONED RESEARCH

Commissioned Data Science Analysis
Commissioned Supervision



CONSULTATIONS

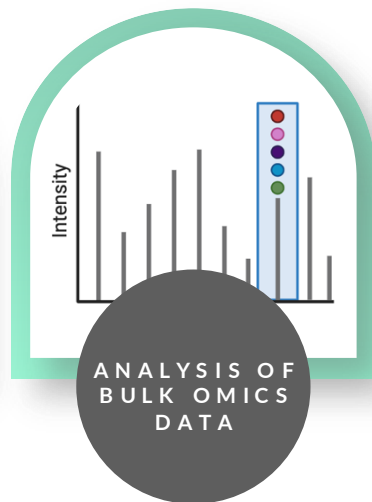
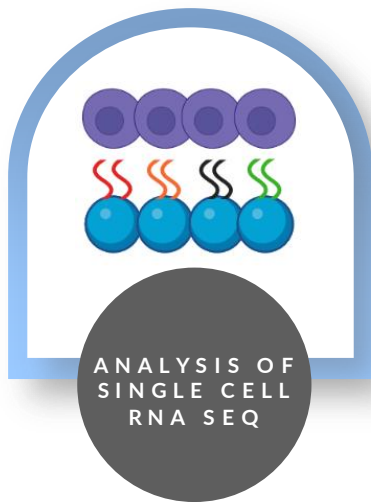
Need guidance? Drop by for a
consultation on your research
project

MATCHMAKING

Conference, seminars and
networking events - Join us!



Commissioned Research



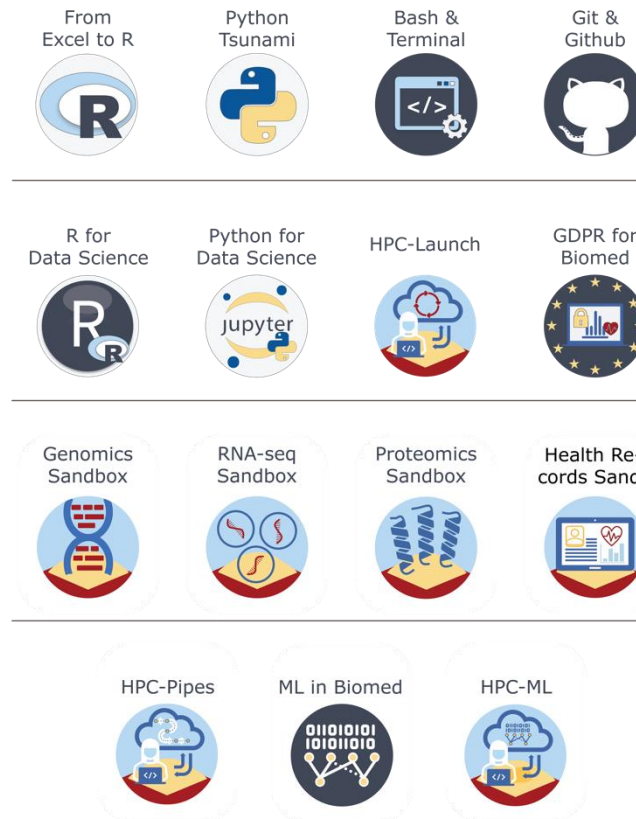
- Machine Learning
- Biometric Data
- Metagenomics
- Epigenetics
- Biological Networks
- Seq. Motif Discovery
- Text mining
- *Biostatistical Models*

HeaDS Courses



Skill Sets at Different Levels:

- Programming (R, Python)
- Omics Data Analysis
- Technical skills (Bash, Git)
- High Performance Compute Tools
- Machine Learning for HDS



From Excel to R

The next **From Excel to R** course

- **May 29 & June 1**
- Registration deadline **1 MAY**
- Scan the QR to find the course calendar

<https://heads.ku.dk/calendar>



The Python Tsunami Team



The Team:

Thilde Terkelsen

Valentina Sora

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Iñigo Prada

Alba Martínez



Original Content:

Henrike Zschach (Research IT)
Alberto Delgado (DTU, Biosustain),
Henry Webel (DTU, Biosustain),
Annalaura Nielsen (University Hospital),
Rita Colaço (PRI)

About this course

Day 1		Day 2		Day 3	
08:45-09:00	Morning Coffee	08:45-9:00	Morning Coffee	08:45-09:00	Morning Coffee
09:00-09:30	Intro to Python & Colab	09:00-10:30	Pandas: Series, Dataframes, Indexing	09:00-09:45	Recap quiz + Q&A
09:30-10:10	Variables & Data Types			09:45-10:30	Visualization
10:10-10:20	Break	10:30-10:45	Break	10:30-10:45	Break
10:20-11:00	Iterables I: Lists	10:45-11:30	Pandas: Data selection & Summary	10:45-12:00	Visualization
11:00-11:10	Break				
11:10-12:00	Iterables II: Sets, Dicts & Tuples	11:30-12:00	Pandas: Modifying Data		
12:00-13:00	Lunch	12:00-13:00	Lunch	12:00-13:00	Lunch
13:00-14:15	Booleans, Operators & Conditions	13:00-14:00	Pandas: Groupby, Sort & Impute	13:00-13:30	Local Installations
14:15-14:30	Break	14:00-14:15	Break	13:30-16:00	Dataset Exercise
14:30-15:30	Loops	14:15-14:45	Pandas: Groupby, Sort & Impute		
15:30-16:00	Functions	14:45-15:00	Pandas Q&A		
16:00	See you tomorrow!	15:00	See you tomorrow!	16:00	Course Finished!



Python

Why Python?

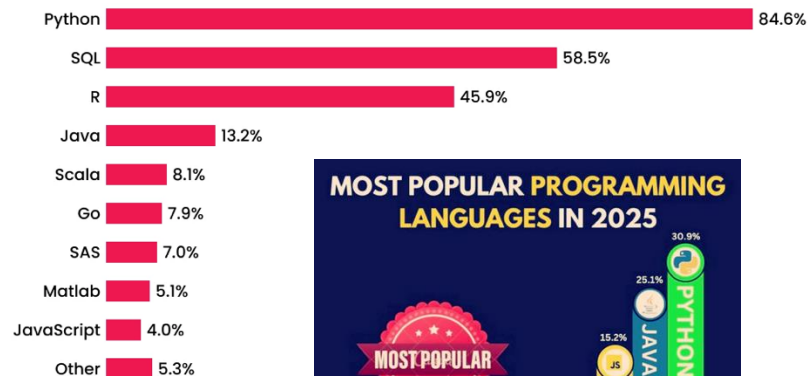


Python is a great programming language for both beginners and advanced programmers:

- Easy to grasp, close to natural language
- Many learning resources available
- Large community
- Libraries / Packages
- Makes machine learning 'easy'

Required Programming Languages

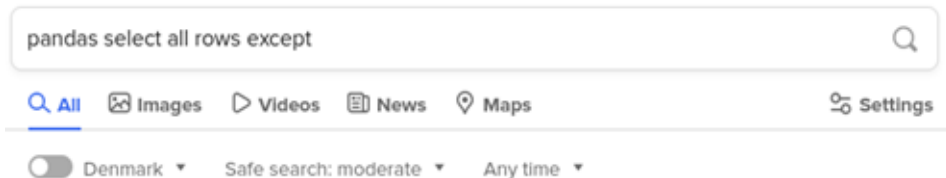
Data Scientists 2025



Where to get help



Online communities such as **stackoverflow** are important tools in programming.

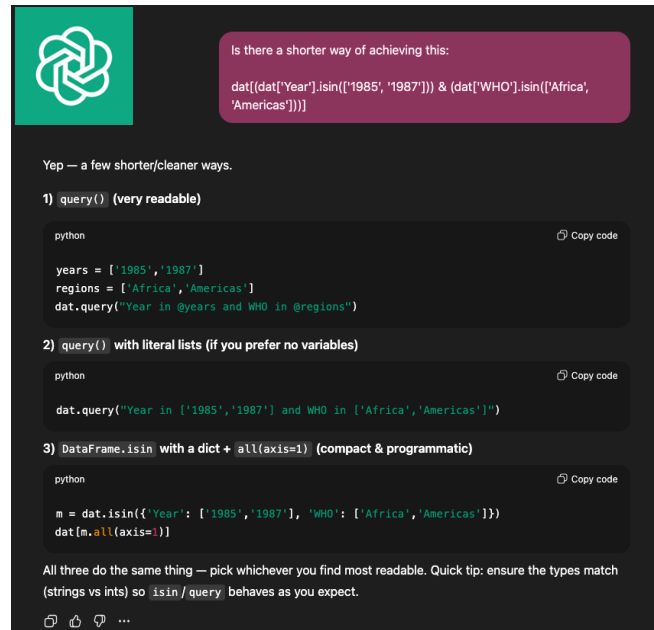


<https://stackoverflow.com/questions/28256761/select-pandas-rows-by-excluding-index-num...>

select pandas rows by excluding index number - Stack Overflow

I'm looking to slice a **Pandas** dataframe by using index numbers. I have a list/core index with the index numbers that i do NOT need, shown below **pandas**. Stack Overflow ... Use a list of values to **select rows** from a **Pandas** dataframe. 2015. Delete a column from a **Pandas** DataFrame. 1290. How to drop **rows** of **Pandas** DataFrame whose value in a certain ...

Or ask **ChatGPT**. But remember: “With great power comes great responsibility!”

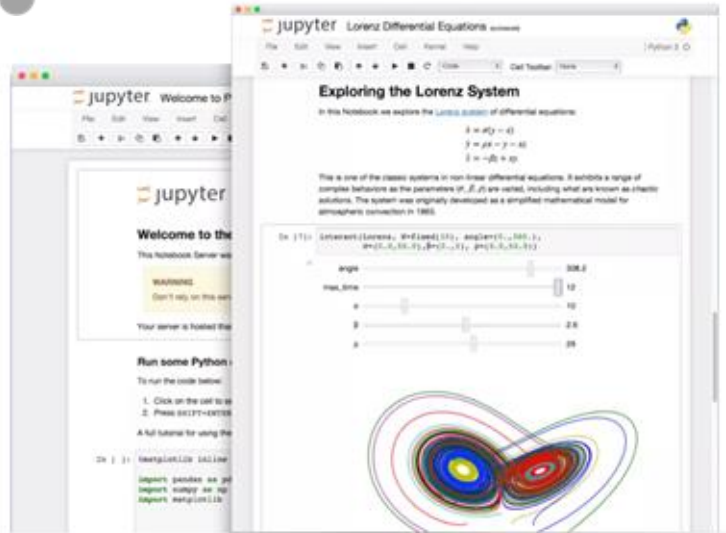


Python Environments

Jupyter notebook



Jupyter Notebook is an **open-source application** to create and share documents that contain code, visualizations and text (markdown).



- Browser-based development environment for creating, running and sharing Python code
- Combine code with text and output – generates; html, pdf, website
- Runs from your **local installation**. I.e. you need Python and the libraries you want to use installed on your computer

Jupyter notebook



variables.ipynb Save in GitHub to keep changes

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text Run all Copy to Drive

Python Tsunami

Variables and Data Types

Variable Assignment

Text cell →

Code cell →

```
> A variable is simply a name that holds a value.

* List item
* List item

|

> Variable **containers** --> name and value

* Create variables by assigning a value to a name (just like using variables in math).
* Variable names should be meaningful, i.e. not just 'a', 'b', 'c'
* Variables are always **assigned** with the variable name on the left and the value on the right of the **equals** sign. For instance:

    * 'a_variable = 100'
    * assigned to other variables: 'another_variable = a_variable'
    * reassigned at any time: 'a_variable = 435'
    * assigning several variables at the same time: 'all, at, once = 1, 130, 43'

* Variables **must** be assigned before they can be used.
```

```
[1]
# Create a variable x by assigning a value to x
x = 100
print(x)

# What is the container and what are the data here?
```

A variable is simply a name that holds a value.

- List item
- List item

Variable **containers** --> name and value

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 - `a_variable = 100`
 - assigned to other variables: `another_variable = a_variable`
 - reassigned at any time: `a_variable = 435`
 - assigning several variables at the same time: `all, at, once = 1, 130, 43`
- Variables **must** be assigned before they can be used.

Jupyter notebook



Getting help inside the notebook

```
[5] ✓ Os help(print)
... Help on built-in function print in module builtins:

print(*args, sep=' ', end='\n', file=None, flush=False)
    Prints the values to a stream, or to sys.stdout by default.

    sep
        string inserted between values, default a space.
    end
        string appended after the last value, default a newline.
    file
        a file-like object (stream); defaults to the current sys.stdout.
    flush
        whether to forcibly flush the stream.
```

If you don't like
AI to do the
work for you
(Colab specific)

Settings

Site	>	☒ Show AI-powered inline completions
Editor	>	☐ Consented to use generative AI features
AI assistance	>	☐ Hide generative AI features
Colab Pro	>	Gemini can make mistakes, so double-check responses and use code with caution .
GitHub	>	
Miscellaneous	>	



Google Colab is a Jupyter Notebook hosted on Google's servers, not your own machine. It still runs in your browser.

- Tool to write, execute and share Python code through the browser
- Requires no setup to use and provides free access to computing resources on Google's servers including GPUs
- Connected to a Google account and data and notebooks can be accessed through Google Drive.

We'll use Colab during the course.

Python Libraries

Python has many libraries, also called packages, that other programmers have developed. Find and **use** them!

Well-maintained libraries generally are:

- Tested
- Optimized
- Documented

During this course we will use:

- Pandas (all the data analysis)
- Math (basic math)
- Seaborn (visualization)



Running Python from a local installation requires you to have libraries **downloaded & installed** before you can use them.

On **Google Colab** you can generally just import, they are already installed.

- Import the math library:

```
import math
```

- Now I can use functions from that library, i.e. calculating the logarithm or square root:

```
math.log(3)
```

```
math.sqrt(4)
```

Course Material



Center-for-Health-Data-Science / PythonTsunami Public

forked from [pythontsunami/teaching](#)

[Code](#) [Issues](#) 10 [Pull requests](#) [Discussions](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

spring2022 16 branches 0 tags

[Go to file](#) [Add file](#) [Code](#)

This branch is 111 commits ahead of [pythontsunami/teaching:heads](#). [Contribute](#) [Fetch upstream](#)

hezscha Add files via upload 078c67a 2 days ago 376 commits

Conditionals	Add files via upload	6 days ago
Exercise	Add files via upload	2 days ago
Functions	Add files via upload	2 days ago
Introduction_and_tools	Add files via upload	7 days ago
Iterables	Add files via upload	2 days ago
Loops	Add files via upload	2 days ago
Pandas	Minor changes to pandas examples	6 days ago
Recap	Add files via upload	9 days ago
Variables_data_types	Add files via upload	6 days ago

You can find the course material here:

<https://github.com/Center-for-Health-Data-Science/PythonTsunami>

Course Material



spring2022 PythonTsunami / Variables_data_types / [Go to file](#) [Add file](#) [...](#)

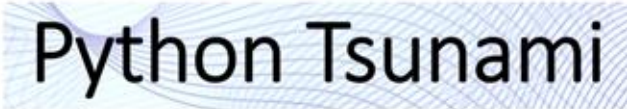
This branch is 111 commits ahead of pythontsunami/teaching:heads. [Contribute](#) [Fetch upstream](#)

hezscha Add files via upload ab1c127 6 days ago [History](#)

..

README.md	Minor fixes	7 days ago
variables.ipynb	Update Colab link within the notebook	6 days ago
variables_solutions.ipynb	Add files via upload	6 days ago

README.md

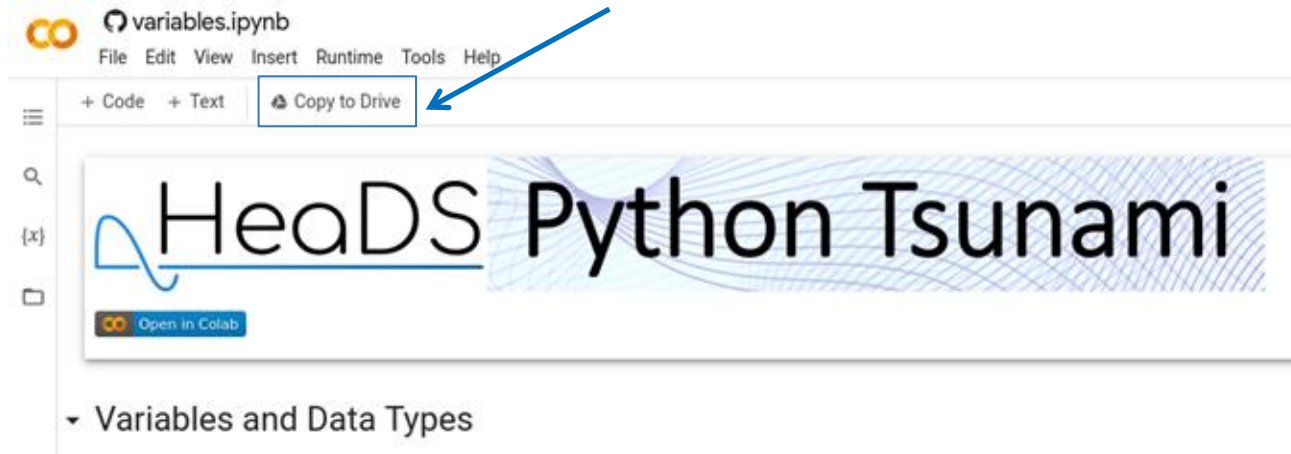
notebook	content
variables.ipynb Open in Colab	Variables and data types



Course Material



Remember to **save a copy** to your own google drive so you can save your notes and exercises!



Short Introduction



Take the next 5 mins to introduce yourself at your table:

- Name
- Position/Unit
- What you do (very briefly!)
- Why would you like to learn Python?